Markus Trunschke

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Date of birth: 20 July 1993
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Research and Teaching Interests

Economics of Innovation, Empirical Industrial Organization, Economics of Digitization, Applied Microeconometrics, Computational Economics, Structural Modelling

Current Positions and Affiliations

5.2024-present	**Postdoctoral researcher** at the Hoover Institution, Stanford University
12.2023-present	**Voluntary Research Associate** at KU Leuven Department of Managerial Economics, Strategy and Innovation (MSI)
4.2024-present	**Junior Research Associate** at ZEW-Mannheim, Mannheim Department of Innovation Economics and Industrial Dynamics

Past Positions and Affiliations

10.2017-4.2024	**Researcher** at ZEW-Mannheim, Mannheim Department of Innovation Economics and Industrial Dynamics
11.-12.2023	**Visiting Scholar** at Hoover Institution, Stanford University, invited by Kenneth L. Judd
8.2022	**Visiting Scholar** at Hoover Institution, Stanford University, invited by Kenneth L. Judd
01.2022-05.2022	**Visiting Scholar** at Pennsylvania State University Department of Economics, invited by Prof. Mark Roberts
8.2021	**Visiting Scholar** at University of Ghent Department of Marketing, Innovation and Organisation, invited by Prof. Maikel Pellens
2019	**External Lecturer** at Baden-Wuerttemberg Cooperative State University, Advanced Microeconomics (undergraduate level)
2015-2016	**Research Assistant** at GESIS - Leibniz-Institute for the Social Sciences

Education

2017-2023	**PhD in Business Economics** at KU Leuven, Department of Managerial Economics, Strategy and Innovation (MSI) Supervisors: Dirk Czarnitzki and Bettina Peters
2019-2020	**Visiting Student** at University of Mannheim Department of Economics
2016-2017	**Master of Science in Economics** at KU Leuven Major: Research in Economics (cum laude) Advanced Econometrics, Structural Modelling, Dynamic Methods, Game Theory, Industrial Organization
2013-2016	**Bachelor of Science in Economics** at University of Mannheim Economic Theory, Microeconomics, Econometrics, Statistics

Research

Publications in Peer-Reviewed Journals

- Pandemic Effects: Do Innovation Activities of Firms Suffer from Long-COVID?, *with Bettina Peters, Dirk Czarnitzki, and Christian Rammer*, Research Policy, Link

Discussion Papers and Research in Progress

- Low-Rank Demand Estimation in High-Dimensional Product Spaces **(Job Market Paper)**, Link
- Production Function Estimation with Strategic Interactions in Output Markets, Link
- Estimating Gross Output Production Functions, *with Kenneth L. Judd*, NBER Working Paper 33205, Link
- Carbon Pricing and Innovation: The Impact of the European Carbon Trading System, Link
- Choosing Technologies: Benefits of Developing Fourth Industrial Revolution Technologies, *with Bettina Peters*, Link
- Efficient Estimation of Likelihood Ratio Confidence Sets, *with Kenneth L. Judd, Gregor Reich, and Carlos Rangel*
- Competition from China in the Product and Input Markets and Innovation in Germany, *with Stephen Ayerst, Philipp Böing, Loren Brandt, and Pierre Mohnen*
- R&D Tax Credits and Directed Technological Change, *with Bernhard Ganglmair*
- Measuring Organisation Capital at the Firm Level: A Production Function Approach, *with Christian Rammer and Felix Roth*, ZEW Discussion Paper
- Industry 4.0 Related Innovation and Firm Growth, *with Vanessa Behrens*, ZEW Discussion Paper

Contract Research and Projects

- Focus Study on Technology markets (Schwerpunktstudie Technologiemärkte), 2023, Link, *Commission of Experts for Research and Innovation*

- Growth Welfare Innovation and Productivity (GROWINPRO),
  European Commission
- Mannheim Innovation Panel: Innovation Activities of German Enterprises,
  German Ministry for Education and Research
- Swiss Report on Research and Innovation, 2018,
  Swiss State Secretariat for Education, Research, and Innovation
- Investments in New Forms of Knowledge-Based Capital (INFOWIK),
  German Ministry for Education and Research.
- Innovation Activity of Companies in East Germany since Reunification (Innovationstätigkeit der Unternehmen in Ostdeutschland seit der Wiedervereinigung),
  Commission of Experts for Research and Innovation
- Development of Research and Development Expenditure in Germany in an International Comparison (Entwicklung der Forschungs- und Entwicklungsausgaben in Deutschland im internationalen Vergleich), *KfW - Credit Institute for Reconstruction*

Software

- **TJProdEst.jl** - Julia implementation of my own gross-output production function estimator developed with Kenneth L. Judd (NBER Working Paper 33205). Link
- **GNRProdEst.jl** - Julia implementation of the Gandhi, Navarro, and Rivers (2020) gross-output production function estimator. Registered in the Julia General registry. Link
- **ACFProdEst** - Julia implementation of the Ackerberg, Caves, and Frazer (2015) production function estimator. Link

Dissemination

- Coverage of the *Pandemic Effects* study in **WirtschaftsWoche** (14.4.2023), Link
- **WIPO Global Innovation Index 2022**, invited expert chapter: Productivity impact of the Fourth Industrial Revolution, *with Bettina Peters*, Link

Teaching

- **Advanced Microeconomics** (undergraduate), External Lecturer, Baden-Wuerttemberg Cooperative State University, 2019

Supervision and Mentoring

- Co-supervision of two Master's theses and one Bachelor's thesis with Prof. Dr. Bernhard Ganglmair, University of Mannheim, 2021-2024
- YES! - Young Economic Summit, 2018-2019, leading student teams in preparation and competition

Conference and Seminar Organisation

- Internal seminar series with internal and external speakers, Department of Innovation Economics and Industrial Dynamics, ZEW-Mannheim, 2022-2024
- 9th ZEW/MaCCI Conference on the Economics of Innovation and Patenting (Mannheim, Germany), 2022
- 8th ZEW/MaCCI Conference on the Economics of Innovation and Patenting (Mannheim, Germany), 2019

Committees

- Scientific Committee, 11th ZEW/MaCCI Conference on the Economics of Innovation and Patenting (Mannheim, Germany), 2026

Memberships of Scientific Societies

- MaCCI - Mannheim Centre for Competition and Innovation (joint centre of ZEW-Mannheim and the University of Mannheim), 2017-2024
- European Economic Association, 2021, 2023, 2026

Invited Seminar Presentations

- Quantitative Economic Policy Seminar, Faculty of Economics and Business, Goethe University Frankfurt, 2025
- MSI Seminar, MSI, KU Leuven, 2025
- CIS seminar series, BETA, University of Strasbourg, 2023

Research Presentations

- International Industrial Organization Conference (Philadelphia, USA), 2025
- Applied Micro Workshop, Pennsylvania State University (State College, PA, USA), 2025
- IO Student Workshop, Stanford University (Stanford, CA, USA), 2025
- PRICE Seminar, ZEW-Mannheim (Mannheim, Germany), 2024

- Comparative Analysis of Enterprise Data (CAED) (State College, USA), 2024
- 50th EARIE Annual Conference (Rome, Italy), 2023
- 28th Annual Conference of the European Association of Environmental and Resource Economists (EAERE) (Limassol, Cyprus), 2023
- DRUID Conference (Lisbon, Portugal), 2023
- 19th CEPR/JIE School on Applied Industrial Organization (Cambridge, England), 2023
- Poster presentation at Munich Summer Institute (Munich, Germany), 2023
- 11th Mannheim Conference on Energy and the Environment (Mannheim, Germany), 2023
- 9th ZEW/MaCCI Conference on the Economics of Innovation and Patenting (Mannheim, Germany), 2022
- Munich Summer Institute Ph.D Workshop (Munich, Germany), 2022
- Eu-SPRI conference (Utrecht, Netherlands), 2022
- International Industrial Organization Conference (Boston, USA), 2022
- IP and Innovation Virtual Seminar (online), 2022
- Applied Microeconomics Seminar Pennsylvania State University (State College,USA), 2022
- CONCORDi Conference (online), 2021
- 17th CAED Conference (Coimbra, Portugal), 2021
- DRUID Conference (Copenhagen, Denmark), 2021
- 16th European Policy for Intellectual Property Conference (EPIP) - PhD Workshop (Madrid, Spain), 2021
- 48th EARIE Annual Conference (Bergen, Norway), 2021
- 18th ISS - International Schumpeter Society Conference (Rome, Italy (online)), 2021
- 9th KID 2021 Summer School on Knowledge Dynamics, Industry Evolution, Economic Development (Nice, France), 2021
- MaCCI IO Day at University of Mannheim (Mannheim, Germany (online)), 2021
- ZEW - Joint PhD Workshop on Innovation, Entrepreneurship and Competition (online), 2020
- 8th PhD Workshop Economics of Innovation, Complexity and Knowledge (Turin, Italy (online)), 2020
- CISS, 9th Competition and Innovation Summer School (Ulcinj, Montenegro), 2019
- CISS, 8th Competition and Innovation Summer School (Ulcinj, Montenegro), 2018

Referee for Journals

International Journal of Industrial Organization (IJIO), Research Policy, Journal of Economic Behavior and Organization, European Journal of Political Economy, Industry and Innovation, Eurasian Business Review

Languages

German (native), English (C2), Spanish (B1), Dutch (A2, ongoing)

References

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